SHAHAZAD ABDULLA

Robotics Systems Engineering | Autonomous Systems | Embedded & Edge AI

Kottayam, Kerala, India
+91 7736652034 | shahazad.abdulla.engineer@gmail.com | linkedin.com/in/shahazad-abdulla |
github.com/ShahazadAbdulla

Summary

Robotics and Automation Engineering undergraduate focused on **robotics systems engineering**, integrating embedded systems, real-time communication, control, perception, and autonomous navigation. Hands-on experience spanning ROS 2, embedded firmware, sensor fusion, robot simulation, and edge AI, with contributions to open-source robotics software and applied robotics research.

Education

Saintgits College of Engineering (Autonomous) — Kottayam, Kerala, India
Bachelor of Technology in Robotics and Automation Engineering — Expected July 2027
Minor in Embedded Systems and IoT

Skills

- **Programming Languages:** C++, C, Python, Rust, Kotlin, MATLAB, Bash
- **Control Systems & Algorithms:** Ros2 control, PID Control, Extended Kalman Filters (EKF), Visual-Inertial SLAM, Sensor Fusion, Path Planning (Nav2), Kinematics
- **Mechatronics & Hardware:** NVIDIA Jetson, Raspberry Pi, STM32, Arduino, ESP32, RISC-V, DACs, IMUs, LiDAR, RADAR, CAN Bus, MQTT, I2C, SPI, UART/Serial
- **Robotics Software & Simulation:** ROS 2, Gazebo, Isaac Sim, PyBullet, CARLA, Nav2, SLAM Toolbox, micro-ROS, URDF/SDF/XACRO, Digital Twins
- **Embedded Systems & Development Tools:** Zephyr RTOS, FreeRTOS, PlatformIO, ESP-IDF, Docker, Git/GitHub, KiCad
- **Computer Vision & ML:** OpenCV, PyTorch, TensorFlow/Keras, ONNX, TensorRT, YOLO, UFLD, Transformers, Camera Calibration

Professional & Technical Experience

National Taiwan Ocean University (NTOU) — Keelung, Taiwan
TEEP Research Intern, Mechanical and Mechatronic Engineering — Aug 2026 – Present

- Developing a robotics system integrating NVIDIA Jetson Orin Nano, ROS 2 Jazzy, embedded control hardware, containerized software, and a Gazebo Harmonic digital twin of an Ackermann-steered robot.
- Designed a custom KiCad PCB for the STM32F446RE controller, integrated steering control with ROS 2 using Zenbedded and Eclipse Zenoh, and developed Android-to-robot camera, IMU, and GPS streaming over Zenoh.

CSIR - National Aerospace Laboratories (NAL) — Bengaluru, India
Research Intern, Project Orion — June 2025 – July 2025

- Developed a multi-sensor state-estimation and Visual SLAM pipeline for an autonomous inspection UGV, integrating visual-inertial data with Extended Kalman Filter (EKF)-based sensor fusion.
- Developed a data-driven calibration methodology (SBKC) to address IMU noise and kinematic mismatches, reducing end-of-path trajectory error by over 90% compared with a VIO-only baseline.
- Integrated low-level C++ drivers with ROS 2 perception and state-estimation components on NVIDIA Jetson Orin Nano; co-authored research on automated aircraft-structure inspection using 2D/3D computer vision, machine learning, and robotics, published in the ARMS 2025 proceedings.

Team Equinox (BAJA SAEINDIA) — Kottayam, Kerala, India
Captain and Head of Autonomous Systems — Mar 2024 – Present

- Progressed from Autonomous Systems Integration Engineer to Vice Captain and Captain of a 40-member team, achieving **All India Rank (AIR) 7** at aBAJA 2025 and **AIR 11** at aBAJA 2026 while preparing for aBAJA 2027.

- Architected a distributed vehicle-control system connecting NVIDIA Jetson and embedded microcontrollers over CAN, integrating lane keeping, adaptive cruise control, emergency braking, path following, and traffic light and sign detection with a focus on system safety and validation.

Intel Corporation (Intel Unnati Program) Kottayam, Kerala, India
Industrial Trainee, Quality Control Automation May 2025 – July 2025

- Developed a ROS 2-based quality inspection system integrating a PyBullet digital twin, monitoring interface, SQLite logging, OCR (EasyOCR), QR-code decoding, and a custom-trained TensorFlow/Keras model for automated product-label inspection.

Independent Robotics Systems Engineer (Freelance) India / Remote
Robotics, Embedded Systems and Edge AI for Startup Clients Jan 2025 – Jul 2026

- Developed robotics systems for startup clients, including AMRs with pick-and-place capabilities, semi-humanoid prototypes, and embedded projects; integrated ROS 2 with robot hardware, debugged drivers and sensor-actuator interfaces, and deployed TensorRT-based perception on NVIDIA Jetson.

Taiwan–India Smart Cities Project (Saintgits & CCU) India
Lead IoT Systems Engineer Aug 2025 – Present

- Led the development, prototyping, testing, and debugging of ESP32-based IoT sensor nodes that acquire environmental data and transmit it to a central server via MQTT, supporting a cross-institutional federated AI platform.

Open Source Contributions

Zenbedded (ros2_control) Open Robotics
Core Developer and Organization Member Mar 2026 – Present

- Contributing to an agent-less embedded integration stack connecting Zephyr-based microcontrollers with `ros2_control` over Eclipse Zenoh, including transport abstractions and joint-state and command communication.
- Developed zero-allocation, static-offset CDR serialization and contributed to embedded discovery, CI, and latency benchmarking; investigated a Zenoh-Pico unicast keep-alive teardown race.

ROS 2 Contributor and Tutorial Party Tester Open Robotics
Community Contributor May 2025 – May 2026

- Authored merged contributions to `control_msgs` and `ros_gz_bridge`, introducing `JointCommand.msg`, fixing timestamp propagation, and adding configurable `frame_id` support.
- Ranked **3rd globally** in 2025 and **4th globally** in 2026, closing over 230 issues through ROS 2 release testing and tutorial validation.

MPU6050 Library for Aries RISC-V Boards May 2025
Creator and Maintainer

- Developed an open-source C++ sensor library with Kalman-filtered roll and pitch estimation for MPU6050 sensors on Vega/Aries RISC-V platforms.

Selected Technical Projects

- **Quadruped Robotics and Autonomous Navigation:** Migrated Hiwonder PuppyPi to ROS 2 Humble with Nav2, SLAM Toolbox, RF2O, and EKF-based sensor fusion. Implemented remote teleoperation, addressed servo stiction and driver issues, and explored NASA JPL's ROSA agent.
- **Edge AI and Autonomous Perception:** Deployed UFLD on NVIDIA Jetson Orin Nano using TensorRT FP16, achieving 19 ms inference latency and reducing GPU utilisation from 60% to 20%.
- **Self-Balancing Robot and Embedded Control:** Implemented a C++ control system on RISC-V, combining Kalman-filter-based state estimation with PID control for real-time stabilisation.

Leadership & Achievements

- **Robotics and Open-Source Leadership:** Founder and Lead, SG-Robotics Open Source; Lead, ROS Club Saintgits; Vice Chairman, The Robotics Society Saintgits Chapter; Co-Lead and Hardware Mentor, TinkerHub Saintgits, mentoring 200+ students.
- **e-Yantra Robotics Competition 2025–26:** Top 100 national qualifier, Holo Battalion (Team eYRC#4075); received a hardware grant.